

一. 填空题

1.

n

$$\begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$$

$$\mathbb{R}[x]_3 \text{ 或 } \text{span}\{x, x^2\}$$

~~而 $\mathbb{R}[x]_3$ 是 $\mathbb{R}[x]$ 的子空间~~

2.

$$\text{不变因子为 } d_1(\lambda) = 1, d_2(\lambda) = 1$$

$$d_3(\lambda) = (\lambda - 4)(\lambda^2 - 9)$$

不是 $d_1(\lambda), d_2(\lambda)$

$d_3(\lambda)$ 也可以

初等因子为

$$\lambda - 4, \lambda + 3, \lambda - 3$$

3.

$$\frac{t}{4 + \sqrt{2}}$$

$$e^{24}$$

4.

$$\begin{bmatrix} 2 & e^{-t} \\ -\sin t & 0 \end{bmatrix}$$

$$\begin{bmatrix} x^2 & e^{-x} \\ \sin x & x \end{bmatrix}$$

二.

$$A = \begin{bmatrix} 2 & 2 & 1 \\ -2 & 6 & 1 \\ 0 & 0 & 4 \end{bmatrix}$$

A 的 Jordan 标准形为 $\begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & 4 \end{bmatrix}$ 或 $\begin{bmatrix} 4 & 1 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$

二者均可

因此 A 的最小多项式为 $(\lambda - 4)^2$

$$A = P J_A P^{-1}$$

$$f(A) = P f(J) P^{-1} = \begin{bmatrix} f(4) - 2f'(4) & 2f'(4) & f'(4) \\ -2f'(4) & f(4) + 2f'(4) & f'(4) \\ 0 & 0 & f(4) \end{bmatrix}$$

不管用哪种形式，只要算出 $f(A)$ 具有何种形式即可。

$$\cos \frac{\pi}{2} A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_{3 \times 3} = I_3$$

$$e^{tA} = e^{4t} \begin{bmatrix} 1 - 2t & 2t & t \\ -2t & 1 + 2t & t \\ 0 & 0 & 1 \end{bmatrix}$$

$$AA^H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

AA^H 特征值为

$$\lambda_1 = \lambda_2 = 1, \lambda_3 = 0$$

$$A \text{ 的奇异值为 } \sigma_1 = \sigma_2 = 1, \Delta = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$(1I - AA^H)X = 0 \rightarrow \eta_1 = [1, 0, 0]^T$$

$$\eta_2 = [0, 1, 0]^T$$

$$(0I - AA^H)X = 0 \rightarrow \eta_3 = [0, 0, 1]^T$$

$$U = [\eta_1, \eta_2, \eta_3], \quad U^H A U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\parallel$
 $U^H A U$

$$U_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$V_1 = A^H U_1 \Delta^{-H}$$

$$V_1 = \begin{bmatrix} -i & 0 \\ 0 & -i \end{bmatrix} = V$$

$$D = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$A = U D V^H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}$$

四.

A 为正规矩阵.

$$|\lambda I - A| = 0 \quad (\lambda^2 + 4)\lambda = 0 \quad \lambda_1 = 0 \quad \lambda_2 = 2i$$

$$\lambda_3 = -2i.$$

$$\lambda_1 = 0 \rightsquigarrow (0I - A)X = 0 \rightsquigarrow \eta_1 = [0, 0, 1]^T$$

$$\lambda_2 = 2i \rightsquigarrow (2iI - A)X = 0 \rightsquigarrow \eta_2 = [\frac{1}{\sqrt{2}}, \frac{i}{\sqrt{2}}, 0]^T$$

$$\lambda_3 = -2i \rightsquigarrow (-2iI - A)X = 0 \rightsquigarrow \eta_3 = [\frac{1}{\sqrt{2}}, -\frac{i}{\sqrt{2}}, 0]^T$$

(η_1, η_2, η_3 为单位特征向量)

只需将上述单位特征向量即可. η_1, η_2, η_3

表示为实矩阵.

$$G_1 = \eta_1 \eta_1^H = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$G_2 = \eta_2 \eta_2^H = \begin{bmatrix} \frac{1}{2} & -\frac{i}{2} & 0 \\ \frac{i}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_3 = \eta_3 \eta_3^H = \begin{bmatrix} \frac{1}{2} & \frac{i}{2} & 0 \\ -\frac{i}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A = 0 \cdot G_1 + 2i G_2 + (-2i) G_3$$

这里并不会有三部分.

缺一块.

五.

$A^H = A$ 为 Hermite 矩阵.

$$|\lambda I - A| = 0 \quad \lambda_1 = 1 \quad \lambda_2 = 2 \quad \lambda_3 = 5$$

$$\lambda_1 = 1: (I - A)X = 0 \rightsquigarrow \eta_1 = [1, 0, 0]^T$$

$$\lambda_2 = 2: (2I - A)X = 0 \rightsquigarrow \eta_2 = [0, i/\sqrt{3}, 1/\sqrt{3}]^T$$

$$\lambda_3 = 5: (5I - A)X = 0 \rightsquigarrow \eta_3 = [0, 1/\sqrt{3}, 1+i/\sqrt{3}]^T$$

只要 η_1, η_2, η_3 线性无关即可.

$$U = [\eta_1, \eta_2, \eta_3] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & i/\sqrt{3} & 1/\sqrt{3} \\ 0 & 1/\sqrt{3} & 1+i/\sqrt{3} \end{bmatrix}$$

$$U^H A U = U^H A U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix} \quad X = U Y \quad \lambda$$

$$f(x) = X^H A X. \text{ 令 } f(x) = \bar{y}_1 y_1 + 2 \bar{y}_2 y_2 + 5 \bar{y}_3 y_3$$

所有 y_i 值均大于零. 所以 $f(x)$ 为正定二次型.

$$\|A\|_2^2 = \max_{X \neq 0} \frac{\|AX\|_2^2}{\|X\|_2^2} = \max_{X \neq 0} \frac{X^H A^H A X}{X^H X} = \rho(A^H A)$$

$$= \rho^2(A) \quad \rho(A) \text{ 为 } A \text{ 的谱半径.}$$

$$\|A\|_2 = \rho(A)$$

或另一种证法:

A 为正规矩阵, 存在酉矩阵 U 使得

$$U^H A U = \Lambda = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$$

$$A = U \Lambda U^H \quad A^H = U \Lambda^H U^H$$

$$A A^H = U \Lambda \Lambda^H U^H = U \begin{bmatrix} |\lambda_1|^2 & & \\ & \ddots & \\ & & |\lambda_n|^2 \end{bmatrix} U^H$$

$$\|A\|_2 = \max_{j=1}^n (\lambda_j(A A^H))^{1/2} = \rho(\lambda_j) = \rho(A).$$

↓
最大者

$$\|A\|_F^2 = \sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2 = \text{Tr}(AA^H)$$

$$= \sum_{i=1}^m \lambda_i(AA^H)$$

$\lambda_i(AA^H)$ 表示 AA^H 的任意一个特征值

$\rho(AA^H)$ 为所有特征值模长最大者.

并且 AA^H 所有特征值均大于或等于零.

$$\forall \rho(AA^H) \leq \text{Tr}(AA^H) = \|A\|_F^2.$$

t.

$$\sum_{k=1}^{\infty} \frac{k}{10^k} x^k \quad \text{收敛半径为 } 10.$$

$$|\lambda I - A| = 0 \quad \lambda_1 = 4 \quad \lambda_2 = -2$$

$$\rho(A) = 4 < 10. \quad \sum \frac{k}{10^k} A^k \text{ 收敛 (且绝对收敛)}$$

$$\sum \frac{k}{10^k} A^k = \sum_{k=1}^{\infty} k B^k, \quad B = A/10$$

$$\sum_{k=1}^{\infty} k x^k = x(1-x)^{-2}$$

$$\sum_{k=1}^{\infty} k B^k = B(I-B)^{-2} \quad B = \begin{bmatrix} 1/10 & 3/10 \\ 3/10 & 1/10 \end{bmatrix}$$

$$I-B = \begin{bmatrix} 9/10 & -3/10 \\ -3/10 & 9/10 \end{bmatrix}$$

$$\sum_{k=1}^{\infty} k B^k = B(I-B)^{-2} = \begin{bmatrix} 35/72 & 5/8 \\ 5/8 & 35/72 \end{bmatrix}$$

11. A 为复数域上矩阵. 存在可逆矩阵 P , 使得

$$P^{-1}AP = J_A$$

$$J_A = \begin{bmatrix} J_{k_1}(\lambda_1) & & \\ & J_{k_2}(\lambda_2) & \\ & & \ddots \\ & & & J_{k_s}(\lambda_s) \end{bmatrix}$$

$$J_{k_i}(\lambda_i) = \begin{bmatrix} \lambda_i & 1 & & \\ & \lambda_i & 1 & \\ & & \ddots & \vdots \\ & & & \lambda_i \end{bmatrix}_{k_i \times k_i}$$

$$J_{k_i}(\lambda_i) = \lambda_i I_{k_i} + \begin{bmatrix} 0 & 1 & & \\ & 0 & 1 & \\ & & \ddots & \vdots \\ & & & 0 \end{bmatrix}_{k_i \times k_i}$$

$$= \lambda_i I_{k_i} + N_{k_i}$$

$$J_A = \begin{bmatrix} \lambda_1 I_{k_1} & & \\ & \lambda_2 I_{k_2} & \\ & & \ddots \\ & & & \lambda_s I_{k_s} \end{bmatrix} + \begin{bmatrix} N_{k_1} & & \\ & N_{k_2} & \\ & & \ddots \\ & & & N_{k_s} \end{bmatrix}$$

$$A = P J_A P^{-1}$$

$$= P \begin{bmatrix} \lambda_1 I_{k_1} & & \\ & \lambda_2 I_{k_2} & \\ & & \ddots \\ & & & \lambda_s I_{k_s} \end{bmatrix} P^{-1} + P \begin{bmatrix} N_{k_1} & & \\ & N_{k_2} & \\ & & \ddots \\ & & & N_{k_s} \end{bmatrix} P^{-1}$$

$$= B + C$$

其中 B 为 n 阶对角化矩阵, C 为幂零矩阵, 且

$$BC = CB.$$